

$$\lim_{k \rightarrow \infty} \lim_{x \rightarrow x_0} f_k(x) = \lim_{x \rightarrow x_0} \lim_{k \rightarrow \infty} f_k(x) ?$$

кр. К : $\exists \{f_k\} \xrightarrow[k \rightarrow \infty]{\mathcal{D}} f \Rightarrow \forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N}:$

$$\forall k > N \quad \forall p \in \mathbb{N} \quad \forall x \in \mathcal{D} =$$

$$|f_{k+p}(x) - f_k(x)| < \varepsilon$$

$$\Rightarrow \lim_{x \rightarrow x_0} |f_{k+p}(x) - f_k(x)| \leq \varepsilon$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ a_{k+p} & & a_k \end{array}$$

$$\Rightarrow |a_{k+p} - a_k| \leq \varepsilon \Rightarrow \{a_k\} - \text{фунд}$$

$$\Rightarrow \exists \lim_{k \rightarrow \infty} a_k = A \quad \text{Д-тб: } \lim_{x \rightarrow x_0} f(x) = A ?$$

Т.к. 1) $\{f_k\} \xrightarrow{\mathcal{D}} f \Rightarrow \forall \varepsilon > 0 \exists N(\varepsilon):$

$$\forall k > N \quad \forall x \in \mathcal{D} \Rightarrow |f_k(x) - f(x)| < \varepsilon$$

2) $\lim_{k \rightarrow \infty} a_k = A \Rightarrow \forall \varepsilon > 0 \exists \tilde{N}(\varepsilon):$

$$\forall k > \tilde{N} \Rightarrow |a_k - A| < \varepsilon$$

$$\exists N^* = \max(N, \tilde{N}) + 1 \Rightarrow \forall x \in \mathcal{D} \Rightarrow$$

$$\begin{cases} |a_{N^*} - A| < \varepsilon \\ |f_{N^*}(x) - f(x)| < \varepsilon \end{cases}$$

$$3) \exists \lim_{x \rightarrow x_0} f_k(x) = a_k \in \mathbb{R} \Rightarrow$$

$$\forall \varepsilon > 0 \quad \exists \delta = \delta(\varepsilon) > 0 : \forall x \in \mathcal{D} :$$

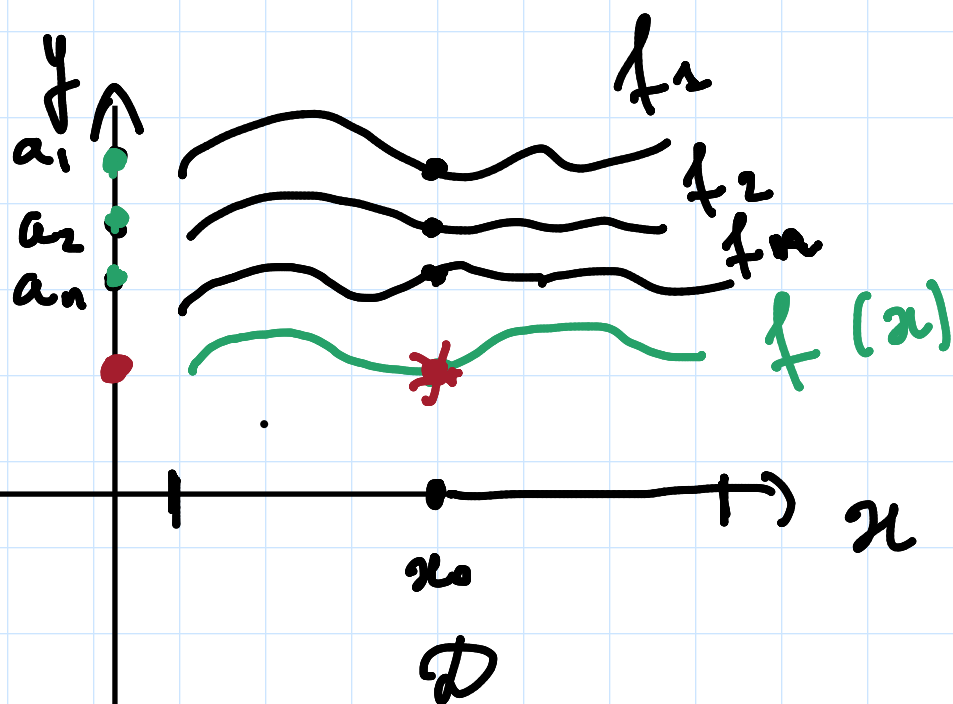
$$0 < |x - x_0| < \delta \Rightarrow |f_{N^*}(x) - a_{N^*}| < \varepsilon$$

$$\Rightarrow \forall x \in \mathcal{V}_\delta^0(x_0) \cap \mathcal{D} :$$

$$|f(x) - A| \leq |f(x) - f_{N^*}(x)| +$$

$$+ |f_{N^*}(x) - a_{N^*}| + |a_{N^*} - A| < \underbrace{\varepsilon}_{< \varepsilon} + \underbrace{\varepsilon}_{< \varepsilon} = 2\varepsilon = \tilde{\varepsilon}$$

△



Ex. $\lim_{k \rightarrow \infty} x^k = \begin{cases} 0, & x \in [0, 1) \\ 1, & x = 1 \end{cases}$

$\mathcal{D} = [0, 1]$

$$\begin{array}{ccc} x^k & \xrightarrow{x \rightarrow 1-0} & 1 \\ \downarrow k \rightarrow \infty & & \downarrow k \rightarrow \infty \\ \begin{bmatrix} 0 \\ 1 \end{bmatrix} & \rightarrow & 0 \neq 1 \end{array}$$

① $\sum_{k=1}^{\infty} f_k(x), f_k(x) : \mathcal{D} \rightarrow \mathbb{R}$

② $\sum_{k=1}^{\infty} f_k(x) \xrightarrow{\mathcal{D}} S(x)$

③ $\forall k \in \mathbb{N} \quad \exists \lim_{x \rightarrow x_0} f_k(x) = a_k \in \mathbb{R}$

$$x_0 - \text{нр.т. } \mathcal{D} \Rightarrow \begin{cases} \sum_{k=1}^{\infty} a_k = A \\ \lim_{x \rightarrow x_0} S(x) = A \end{cases} \text{ num}$$

$$\lim_{x \rightarrow x_0} S(x) = \lim_{x \rightarrow x_0} \sum_{k=1}^{\infty} f_k(x) = \sum_{k=1}^{\infty} \lim_{x \rightarrow x_0} f_k(x) \quad \left. \begin{array}{l} \text{p.c.} \\ \} \notin \mathbb{R} \end{array} \right\}$$

Δ φ-в непрерывно вытекает из (T) что

$$\{f_k(x)\} = \{S_n(x)\} \quad \Delta \quad S_n = \sum_{k=1}^n f_k$$

$$(T) \quad \mathcal{D} \{f_k(x)\}, \quad f_k: \mathcal{D} \rightarrow \mathbb{R}$$

$$\exists f(x), \quad f: \mathcal{D} \rightarrow \mathbb{R}, \quad x_0 \in \mathcal{D}$$

$$(1) \quad \{f_k(x)\} \xrightarrow{\mathcal{D}} f \quad \left| \Rightarrow f(x) \in C^0(x_0) \right.$$

$$(2) \quad f_k(x) \in C^0(x_0), \quad \forall k$$

$$(\text{even } f_k \in C^0(\mathcal{D}) \Rightarrow f \in C^0(\mathcal{D}))$$

$$\Delta \quad 1) \exists x_0 - \text{нр.т. } \mathcal{D} \Rightarrow \text{т.к. } \forall f \in C^0(\text{нр.т. } x_0)$$

$\Rightarrow$ грб. грк-тв.

$$2). \exists x_0 - \text{нр.т. } \mathcal{D} \Rightarrow T! \quad // f_k(x_0)$$

$$\mathcal{D} \quad \lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} \lim_{k \rightarrow \infty} f_k(x) = \lim_{k \rightarrow \infty} \lim_{x \rightarrow x_0} f_k(x) = f(x_0)$$

$$\stackrel{\text{def}}{\Rightarrow} f(x) \in C^0(x_0) \quad \Delta$$

$$(T) \quad \mathcal{D} \quad \sum_{k=1}^{\infty} f_k(x), \quad f_k: \mathcal{D} \rightarrow \mathbb{R}, \quad x_0 \in \mathcal{D}$$

$$\exists (1) \quad \sum_{k=1}^{\infty} f_k(x) \xrightarrow{\mathcal{D}} S(x)$$

$$(2) \quad f_k(x) \in C^0(x_0) \quad \forall k$$

$$(\text{even } \forall f_k \in C^0(\mathcal{D}) \Rightarrow S(x) \in C^0(\mathcal{D}))$$

$$\left| \Rightarrow S(x) \in C^0(x_0) \right.$$

Δ D-то непрерыв. вытекает из ТР что
 $\{S_n(x)\}$. Δ

(T) ∃ $\{f_k(x)\}$, $f_k, f: [a, b] \rightarrow \mathbb{R}$

∃ (1) $f_k \in C^0[a, b] \Rightarrow f \in C^0[a, b]$
 (2) $f_k \xrightarrow{[a, b]} f$

и $\forall x \in [a, b]$

$$\int_a^x f_k(x) dx \xrightarrow{[a, b]} \int_a^x f(x) dx, \quad x \in [a, b] *$$

$$\int f_1, \int f_2, \dots, \int f_n, \dots$$

↓↓

$$\int f$$

Δ ∃ $f_k \xrightarrow{[a, b]} f \stackrel{\text{def}}{=} \forall \varepsilon > 0 \exists N(\varepsilon) : \forall k > N, \forall x \in [a, b]$
 $\Rightarrow |f_k(x) - f(x)| < \varepsilon.$

∃ $k > N$

$$\exists \left| \int_a^x f_k dx - \int_a^x f dx \right| = \left| \int_a^x (f_k - f) dx \right| \leq$$

$$\leq \int_a^x |f_k - f| dx \leq \varepsilon |x - a| \leq \varepsilon (b - a) = \tilde{\varepsilon}$$

$$\forall x \in [a, b]$$

def PC

$\Rightarrow$

*

Δ

(T) ∃ $\sum_{k=1}^{\infty} f_k(x)$, $f_k: [a, b] \rightarrow \mathbb{R}$

∃ (1) $f_k \in C^0[a, b]$

(2) $\sum_{k=1}^{\infty} f_k \xrightarrow{[a, b]} S(x) \Rightarrow S(x) \in C^0[a, b]:$

$$\int_a^x \left(\sum_{k=1}^{\infty} f_k \right) = \sum_{k=1}^{\infty} \left(\int_a^x f_k dx \right) \quad x \in [a, b]$$

Δ ϕ -to $\Rightarrow$ by $T \uparrow$ give $\{S_n(a)\}$ Δ

(T) $\{f_k\}$, $f_k: [a, b] \rightarrow \mathbb{R}$

[(1) $f_k \in C^1 [a, b]$

(2) $\exists x_0 \in [a, b] : \{f_k(x_0)\} = (C_x) \Rightarrow$

(3) $\{f'_k(x)\} \xrightarrow{[a, b]} g(x)$

$$\int f'_k \xrightarrow{[a, b]} f$$

$$f' = g, \quad x \in [a, b], \quad f \in C^1[a, b]$$

$\Delta \stackrel{(1)+(3)}{=} f \in C^0[a, b]$

$$\circ \lim_{k \rightarrow \infty} \int_{x_0}^x f'_k dx = \int_{x_0}^x (\lim f'_k) dx = \int_{x_0}^x g dx \quad \left(\xrightarrow{[a, b]} \right)$$

$$\circ \lim_{k \rightarrow \infty} \int_{x_0}^x f'_k dx = \lim_{k \rightarrow \infty} (f_k(x) - f_k(x_0)) = \int_{x_0}^x g dx$$

by (2), so $\exists \lim_{k \rightarrow \infty} f_k(x_0) = C \Rightarrow$

$$f(x) = \lim_{k \rightarrow \infty} f_k(x_0) + \int_{x_0}^x g dx = C + \int_{x_0}^x g dx \quad \left(\xrightarrow{[a, b]} \right)$$

Термин $\Rightarrow$

$$\circ f'(x) = g(x), \quad x \in [a, b] \quad \Delta$$

① $\sum_{k=1}^{\infty} f_k(x)$, $f_k: [a, b] \rightarrow \mathbb{R}$

$$1 \text{ (I) } f_{\alpha}(x) \in C^1[a, b]$$

$$\textcircled{2} \quad \exists x_0 \in [a, b] : \sum_{k=1}^{\infty} f_k(x_0) = \textcircled{\cos}$$

$$\textcircled{B} \quad \sum_{k=1}^{\infty} f_k'(x) \xrightarrow{[a,b]} \int (x) \quad | \Rightarrow$$

$$\int \sum_{k=1}^{\infty} f_k(x) \xrightarrow{[a,b]} S(x)$$

$$S'(x) = \tilde{S}(x), \quad x \in \mathcal{L}(a, b]$$

$$S \in C^1[a, b].$$

$$\left(\sum_{k=1}^{\infty} f_k(x) \right)' = \sum_{k=1}^{\infty} f_k'(x)$$

$$\Delta \emptyset. \Rightarrow \text{we } T \uparrow \text{ give } \{ S_n(x) \} \Delta$$

Tip $(x^{1/3} - x) + (x^{1/5} - x^{1/3}) + \dots + (x^{\frac{1}{2n+1}} - x^{\frac{1}{2n-1}}) + \dots$

$$S(x) \in C^0(\mathbb{R})?$$